

Qatar-1b

R-2637

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110919 · created 2026-08-29T00:58:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455823.71542 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.63 +/- 2.14 [%]
Orbital Inclination (inc)	88.78 +/- 2.92
Airmass coefficient 1 (a1)	0.053 +/- 0.028
Airmass coefficient 2 (a2)	2.37 +/- 1.0
Scatter in the residuals of the lightcurve fit is	50.22 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	3.0
Transit Duration (day)	0.083 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.056 vs 0.1463 (deviation 0.58σ)
Duration fit vs published	0.083 d vs 0.0692 d (deviation 1.46σ)
Mid-time O–C	-0.9 min
Depth SNR	2.5
Residual scatter	50.22 %

Light curve

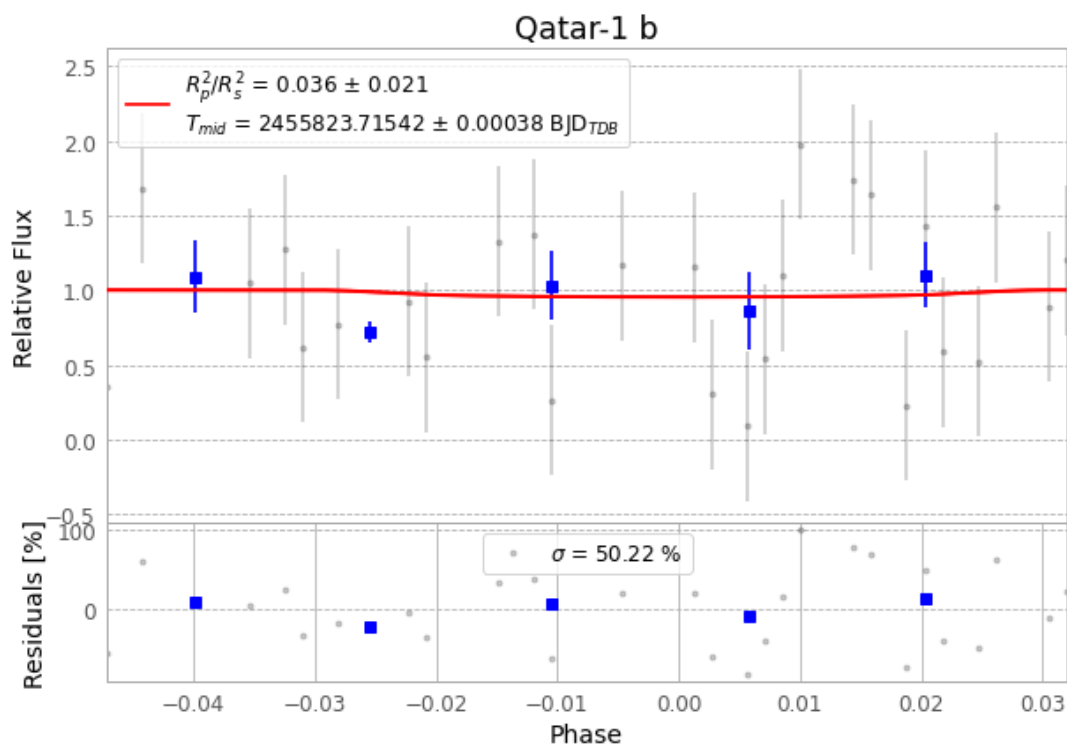


Figure 1 — FinalLightCurve_Qatar-1 b_2011-09-18.png (SHA-256 0bbc162de7e2ce71...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [212, 464], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eee42615d68e0df3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), Qatar-1110919033012.FITS → Qatar-1110919061213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110919.log (789c0e4d7ded...), FinalLightCurve_Qatar-1 b_2011-09-18.png (0bbc162de7e2...), FinalLightCurve_Qatar-1 b_2011-09-18.csv (fa9013252fe0...)

Compute resources

Wall time 97.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:58Z.